Alternating Current

Single Correct Type Questions

1. An alternating is given by the equation $i = i_1 \sin \omega t + i_2 \cos \omega t$. Then rms current will be:

[26 Feb, 2021 (Shift-I)]

(*a*) $\frac{1}{\sqrt{2}}\left(i_1^2 + i_2^2\right)^{\frac{1}{2}}$ (*b*) $\frac{1}{\sqrt{2}}\left(i_1 + i_2\right)^2$

(*c*) $\frac{1}{\sqrt{2}}\left(i_2 + i_2\right)$ (*d*) $\frac{1}{2}\left(i_1^2 + i_2^2\right)^{\frac{1}{2}}$

2. An AC current is given by $I = I_1 \sin\omega t + I_2 \cos\omega t$. A hot wire ammeter will give a reading:

[17 March, 2021 (Shift-I)]

(*a*) $\frac{I_1 + I_2}{2\sqrt{2}}$ (*b*) $\sqrt{\frac{I_1^2 + I_2^2}{2}}$

(*c*) $\frac{I_1 + I_2}{\sqrt{2}}$ (*d*) $\sqrt{\frac{I_1^2 - I_2^2}{2}}$

3. An alternating emf $E = 440 \sin 100\pi t$ is applied to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi} H$. If an a.c. ammeter is connected in the circuit, its reading will be:

[29 July, 2022 (Shift-I)]

(*a*) 4.4 A (*b*) 1.55 A

(*c*) 2.2 A (*d*) 3.11 A

4. The equation of current in a purely inductive circuit is $5\sin\left(49\pi t - 30^\circ\right)$. If the inductance is 30 mH then the equation for the voltage across the inductor, will be:

$\left\{\text{Let } \pi = \frac{22}{7}\right\}$

[28 July, 2022 (Shift-I)]

(*a*) $1.47 \sin (49\pi t - 30°)$

(*b*) $1.47 \sin (49\pi t + 60°)$

(*c*) $23.1 \sin (49\pi t - 30°)$

(*d*) $23.1 \sin (49\pi t + 60°)$

5. An AC source rated 220V, 50Hz is connected to a resistor. The time taken by the current to change from its maximum to the rms value is

[18 March, 2021 (Shift-I)]

(*a*) 2.5 ms

(*b*) 25 ms

(*c*) 2.5 s

(*d*) 0.25 ms

6. A circuit connected to an ac source of emf $e = e_0\sin(100t)$ with t in seconds, gives a phase difference of $\frac{\pi}{4}$ between the emf e and current i. Which of the following circuits will exhibit this? **[8 April, 2019 (Shift-II)]**

(*a*) RC circuit with $R = 1$ kΩ and $C = 1$μF.

(*b*) RL circuit with $R = 1$ kΩ and $L = 1$mH.

(*c*) RL circuit with $R = 1$ kΩ and $L = 10$ mH.

(*d*) RC circuit with $R = 1$ kΩ and $C = 10$ μF.

7. A 0.07 H inductor and a 12 Ω resistor are connected in series to a 220 V, 50 Hz ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take π as 22/7]

[27 July, 2021 (Shift-I)]

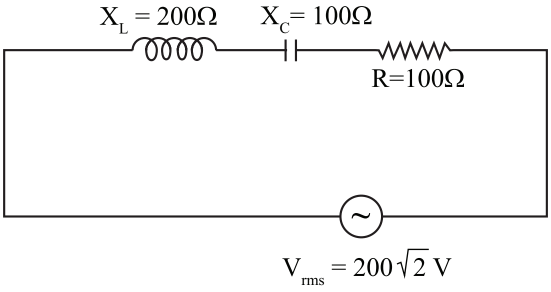
(*a*) 88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

(*b*) 8.8 A and $\tan^{-1}\left(\frac{11}{6}\right)$

(*c*) 0.88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

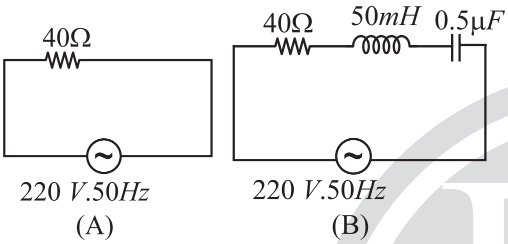
(*d*) 8.8 A and $\tan^{-1}\left(\frac{11}{6}\right)$

8. In the given circuit, rms value of current (I_{rms}) through the resistor R is: **[30 Jan, 2023 (Shift-II)]**



(*a*) 2*A* (*b*) 1/2 A (*c*) 20A (*d*) $2\sqrt{2}$ A

9. For the given figures, choose the correct options: **[29 Jan, 2023 (Shift-II)]**



(*a*) The rms current in circuit (B) can never be larger than that in (A)

(*b*) The rms current in figure (A) is always equal to that in figure (B)

(*c*) The rms current in circuit (B) can be larger than that in (A)

(*d*) At resonance, current in (B) is less than that in (A)

10. A series LCR circuit is connected to a 45 sin (ωt) Volt source. The resonant angular frequency of the circuit is 10^5 rad s^{-1} and current amplitude at resonance is I_0. When the angular frequency of the source is $\omega = 8 \times 10^4$ rad s^{-1}, the current amplitude in the circuit is 0.05 I_0. If $L = 50$ mH, match each entry in List-I with an appropriate value from List-II and choose the correct option. **[JEE Adv, 2023]**

List-I		List-II	
(*p*)	I_0 in mA	(*i*)	44.4
(*q*)	The quality factor of the circuit	(*ii*)	18
(*r*)	The bandwidth of the circuit in rad s^{-1}	(*iii*)	400
(*s*)	The peak power dissipated at resonance in Watt	(*iv*)	2250
		(*v*)	500

(*a*) $p \rightarrow (ii), q \rightarrow (iii), r \rightarrow (v), s \rightarrow (i)$

(*b*) $p \rightarrow (iii), q \rightarrow (i), r \rightarrow (iv), s \rightarrow (ii)$

(*c*) $p \rightarrow (iv), q \rightarrow (v), r \rightarrow (iii), s \rightarrow (i)$

(*d*) $p \rightarrow (iv), q \rightarrow (ii), r \rightarrow (i), s \rightarrow (v)$

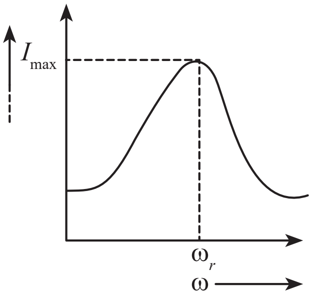
11. For a series LCR circuit, I vs ω curve is shown:

A. To the left of ω_r, the circuit is mainly capacitive.

B. To the left of ω_r, the circuit is mainly inductive.

C. At ω_r, impedance of the circuit is equal to the resistance of the circuit.

D. At ω_r, impedance of the circuit is 0 .

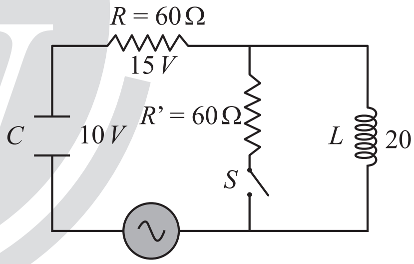


Choose the most appropriate answer from the options given below: **[29 June, 2022 (Shift-I)]**

(*a*) (A) and (*D*) only (*b*) (*B*) and (*D*) only

(*c*) (A) and (*C*) only (*d*) (*B*) and (*C*) only

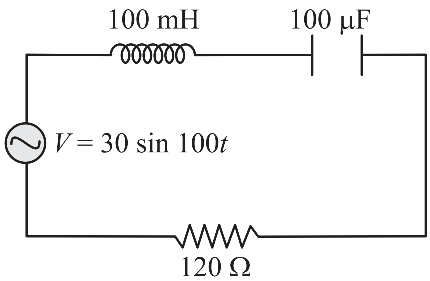
12. The angular frequency of alternating current in a LCR circuit is 100 rad/s. The components connected are shown in the figure. Find the value of inductance of the coil and capacity of condenser. **[25 Feb, 2021 (Shift-I)]**



(*a*) 0.8 H and 150 μF (*b*) 1.33 H and 250μ F

(*c*) 0.8 H and 250 μF (*d*) 1.33 H and 150 μF

13. Find the peak current and resonant frequency of the following circuit (as shown in figure). **[26 Feb, 2021 (Shift-II)]**



(*a*) 2 A and 100 Hz (*b*) 2 A and 50 Hz

(*c*) 0.2 A and 50 Hz (*d*) 0.2 A and 100 Hz

14. A series LCR circuit driven by 300 V at a frequency of 50 Hz contains a resistance $R = 3k\Omega$, an inductor of inductive reactances $X_L = 250\pi\Omega$ and an unknown capacitor. The value of capacitance to maximize the average power should be : (Take $\pi^2 = 10$) **[26 Aug, 2021 (Shift-I)]**

(*a*) $400\mu F$ (*b*) $4\mu F$

(*c*) $40\mu F$ (*d*) $25\mu F$

15. A direct current of 4A and an alternating current of peak value 4A passes through 3 Ω and 2 Ω resistors respectively. Find the ratio of heat generated in two resistance in same interval of time. **[27 July, 2022 (Shift-I)]**

(*a*) 3 : 2 (*b*) 3 : 1 (*c*) 3 : 4 (*d*) 4 : 3

16. An electrical power line, having a total resistance of 2Ω, delivers 1k W at 220 V. The efficiency of the transmission line is approximately: **[05 Sep, 2020 (Shift-I)]**

(*a*) 96% (*b*) 72% (*c*) 91% (*d*) 85%

17. A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60 Ω) is driven by an AC source of 24 V/50 Hz. The energy dissipated in the circuit in 60 s is: **[9 Jan, 2019 (Shift-II)]**

(*a*) 5.65×10^2 J (*b*) 2.26×10^3 J

(*c*) 5.17×10^2 J (*d*) 3.39×10^3 J

18. Match the List-I with List-II.

	List-I		List-II
A.	AC generator	I.	Presence of both L and C
B.	Transformer	II.	Electromagnetic Induction
C.	Resonance phenomenon to occur	III.	Quality factor
D.	Sharpness of resonance	IV.	Mutual Inductance

Choose the correct answer from the options given below: **[1 Feb, 2023 (Shift-I)]**

(*a*) A → IV, B → II, C → I, D → III
(*b*) A → II, B → I, C → III, D → IV
(*c*) A → II, B → IV, C → I, D → III
(*d*) A → IV, B → III, C → I, D → II

19. A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns, given the output power at 230 V. If the current in the primary winding of the transformer is 5A, and its efficiency is 90%, the output current would be **[9 Jan, 2019 (Shift-II)]**

(*a*) 20 A (*b*) 40A

(*c*) 45A (*d*) 25A

Integer Type Questions

20. The alternating current is given by

$$i = \left\{\sqrt{42}\sin\left(\frac{2\pi}{T}t\right) + 10\right\} A$$

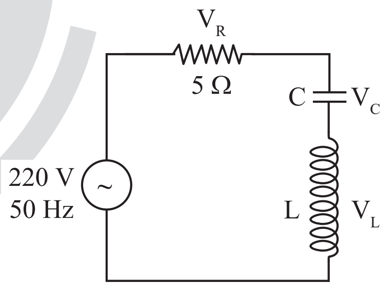
The r.m.s. value of this current is ________ A **[27 Aug, 2021 (Shift-I)]**

21. A series LCR circuit consists of R = 80Ω. $X_L = 100\Omega$, and $X_C = 40\Omega$. The input voltage is 2500 cos(100 πt)V. The amplitude of current, in the circuit, is ________ A. **[31 Jan, 2023 (Shift-II)]**

22. An inductor of inductance 2μH is connected in series with a resistance, a variable capacitor and an AC source of frequency 7 kHz. The value of capacitance for which maximum current is drawn into the circuit is $\frac{1}{x}F$, where the value of x is_______.

(Use $\pi = \frac{22}{7}$) **[29 Jan, 2023 (Shift-II)]**

23. In the given circuit, the magnitude of V_L and V_C are twice that of V_R. Given that f = 50 Hz, the inductance of the coil is $1/K\pi$ mH. The value of K is ________. **[28 June, 2022 (Shift-II)]**



24. The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rad s^{-1} and 232 rad s^{-1}. The value of resistance in the circuit is R = 5Ω. The self inductance in the circuit is _______ mH. **[28 July, 2022 (Shift-I)]**

25. A telegraph line of length 100 km has a capacity of 0.01 μF/km and it carries an alternating current at 0.5 kilo cycle per second. If minimum impedance is required, then the value of the inductance that need to be introduced in series is ________ mH. (if $\pi = \sqrt{10}$) **[28 June, 2022 (Shift-I)]**

26. A resonance circuit having inductance and resistance 2×10^{-4} H and 6.28 Ω respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is________. [$\pi = 3.14$]

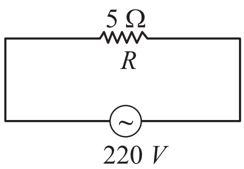
[24 Feb, 2021 (Shift-I)]

27. A series LCR circuit of $R = 5\Omega$, $L = 20\text{mH}$ and $C = 0.5\mu F$ is connected across an AC supply of 250V, having variable frequency. The power dissipated at resonance condition is ________ $\times 10^2$ W.

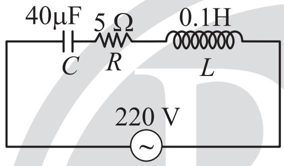
[20 July, 2021 (Shift-II)]

28. Two circuits are shown in the figure (a) & (b). At a frequency of ________ rad /s the average power dissipated in one cycle will be same in both the circuits.

[25 July, 2021 (Shift-II)]



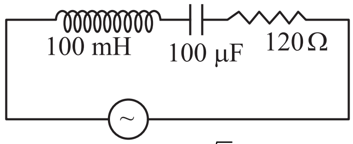
Figure(a)



Figure(b)

29. An AC source is connected to an inductance of 100 mH, a capacitance of 100 μF and a resistance of 120Ω as shown in figure. The time in which the resistance having a thermal capacity 2 J/°C will get heated by 16°C is _____s.

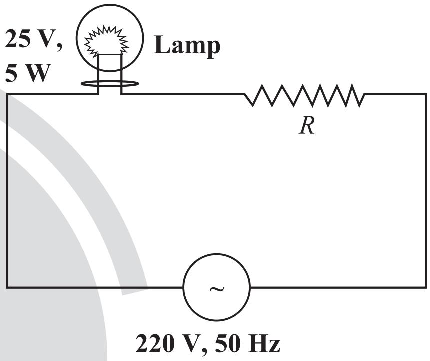
[28 June, 2022 (Shift-I)]



$V = 20\sqrt{2}\sin(100t)$

30. A 220 V, 50 Hz AC source is connected to a 25 V, 5 W lamp and an additional resistance R in series (as shown in figure) to run the lamp at its peak brightness, then the value of R (in ohm) will be _________.

[27 June, 2022 (Shift-I)]



ANSWER KEY

1. (*a*)	**2.** (*b*)	**3.** (*c*)	**4.** (*d*)	**5.** (*a*)	**6.** (*d*)	**7.** (*b*)	**8.** (*a*)	**9.** (*a*)	**10.** (*b*)
11. (*c*)	**12.** (*c*)	**13.** (*c*)	**14.** (*b*)	**15.** (*b*)	**16.** (*a*)	**17.** (*c*)	**18.** (*c*)	**19.** (*c*)	**20.** [11]
21. [25]	**22.** [3872]	**23.** [0]	**24.** [250]	**25.** [100]	**26.** [2000]	**27.** [125]	**28.** [500]	**29.** [15]	**30.** [975]

EXPLANATIONS

1. (*a*) $I = I_1 \sin\omega t + I_2 \cos\omega t$

$I = I_1 \sin\omega t + I_2 \sin(\omega t + 90°)$

$I = \sqrt{I_1^2 + I_2^2}\,\sin(\omega t + \alpha)$

$I_{rms} = \dfrac{I_0}{\sqrt{2}} = \dfrac{\sqrt{I_1^2 + I_2^2}}{\sqrt{2}}$

2. (*b*) A hot wire ammeter reads rms value of current, so

$I = I_1 \sin\omega t + I_2 \cos\omega t$

$= \sqrt{I_1^2 + I_2^2}\ \sin(\omega t + \phi) = I_0 \sin(\omega t + \phi)$, where

$I_0 = \sqrt{I_1^2 + I_2^2}$

$\Rightarrow I_{rms} = \dfrac{I_0}{\sqrt{2}} = \sqrt{\dfrac{I_1^2 + I_2^2}{2}}$

3. (*c*) We have, $E = 440 \sin 100\pi t$

So, $\omega = 100\pi$

$L = \dfrac{\sqrt{2}}{\pi} H$

$X_L = \omega L = 100\sqrt{2}\,\Omega$

$i_{(ms)} = \dfrac{V_{(rms)}}{X_L} = \dfrac{440}{\sqrt{2} \times 100\sqrt{2}} = 2.2A$

4. (*d*) We have, $i = 5 \sin(49\pi t - 30°)$

$L = 30\text{mH},\ \pi = \dfrac{22}{7}$

$X_L = \omega L = 49\pi \times 30 \times 10^{-3} = 462 \times 10^{-2}\ \Omega$

$V_0 = i_0 X_L = 5 \times 462 \times 10^{-2} = 23.1V$

So, $V_L = V_0 \sin(49\pi t - 30° + 90°)$ (Voltage will lead current by 90°)

$\Rightarrow V_L = 23.1 \sin(49\pi t + 60°)$

5. (*a*) $I = I_0 \sin\omega t \Rightarrow \dfrac{I_0}{\sqrt{2}} = I_0 \sin\omega t$

$\Rightarrow \omega t = \dfrac{\pi}{4} \Rightarrow t = \dfrac{\pi}{4\omega} = \dfrac{\pi}{4(2\pi f)}$

$\Rightarrow t = \dfrac{1}{8 \times 30} = \dfrac{1}{400} = 2.5\,\text{ms}$

6. (*d*) Given phase difference $= \dfrac{\pi}{4}$ and $\omega = 100$ rad/s

$\Rightarrow$ Reactance (X) = Resistance (R)

Option (*a*)

$R = 1000\ \Omega$ and $X_C = \dfrac{1}{10^{-6} \times 100} = 10^4\,\Omega$

Option (*b*)

$R = 10^3\ \Omega$ and $X_L = 10^{-3} \times 100 = 10^{-1}\ \Omega$

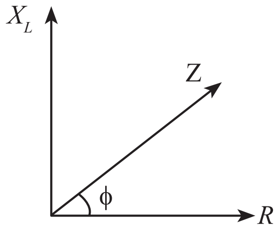
Option (*c*)

$R = 10^3\ \Omega$ and $X_L = 10 \times 10^{-3} \times 100 = 1\Omega$

Option (*d*), $R = 10^3\ \Omega$ and $X_C = \dfrac{1}{10 \times 10^{-6} \times 100} = 10^3\,\Omega$

$\Rightarrow$ Option (*d*) matches the given condition.

7. (*b*) $L = 0.07$ H and $R = 12\ \Omega$, $V_S = 220$ V, 50 Hz



$Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + (2\pi f L)^2}$

$= \sqrt{12^2 + (7\pi)^2} \approx 25\Omega$

Current, $\therefore i = \dfrac{V}{Z} = \dfrac{220V}{25\Omega} = 8.8\,A$

$\tan\phi = \dfrac{X_L}{R} = \dfrac{7\pi}{12} \Rightarrow \phi = \tan^{-1}\left(\dfrac{11}{6}\right)$

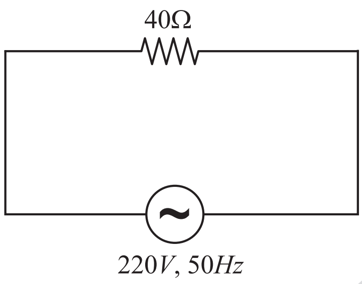
8. (*a*) Independence Z of the circuit is,

$$Z = \sqrt{100^2 + (200-100)^2} = 100\sqrt{2}\Omega$$

$$i_{rms} = \frac{V_{rms}}{Z} = \frac{200\sqrt{2}}{100\sqrt{2}}$$

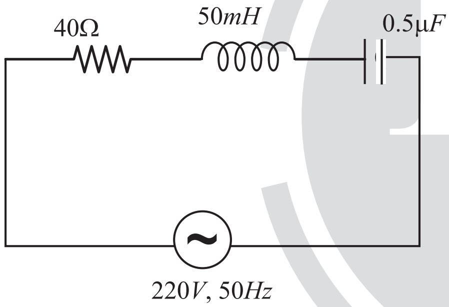
$$\Rightarrow i_{rms} = 2\text{A}$$

9. (*a*) For the circuit in Fig (A)



$$I_{rms} = \frac{220}{40} = 5.5\,A$$

For the circuit in Fig (B)



X_L is not equal to X_C. So rms current in fig (B) can never be larger than the rms current in Fig (A).

10. (*b*) Given, $V = 45 \sin \omega t$, $L = 50$ mH, $\omega_0 = 10^5$ rad/s
$\omega = 8 \times 10^4$ rad/s $= 0.8\ \omega_0$

Resonance frequency of L–C–R circuit,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\therefore C = \frac{1}{L\omega_0^2} = \frac{1}{5\times10^{-2}\times10^{10}}$$

$$= 2 \times 10^{-9}\ F$$

$$I = 0.05 I_0 = \frac{I_0}{20} \Rightarrow Z = 20R$$

$$X_L = \omega L = 8 \times 10^4 \times 5 \times 10^{-2}\Omega = 4\text{k}\Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{8\times10^4\times2\times10^{-9}} = \frac{25}{4}\text{k}\Omega$$

Impedance of the circuit,

$$Z^2 = R^2 + (X_C - X_L)^2$$

$$400R^2 = R^2 + \left(\frac{9}{4}\text{k}\Omega\right)^2$$

$$R = \frac{\frac{9}{4}\text{k}\Omega}{\sqrt{399}} \approx \frac{9}{80}\text{k}\Omega = \frac{900}{8}\Omega$$

Current amplitude,

$$I_0 = \frac{V_0}{R} = \frac{45\times8}{900} = \frac{8}{20}A \approx 0.4A \ = 400\text{ mA}$$

Quality factor,

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{8}{900}\sqrt{\frac{5\times10^{-2}}{2\times10^{-9}}} = \frac{8}{900}\sqrt{25\times10^6}$$

$$= 44.4$$

Also, $Q = \dfrac{\omega_0}{\Delta\omega}$

Therefore bandwidth,

$$\Delta\omega = \frac{\omega_0}{Q} = \frac{10^5}{44.4} = 2250.0$$

Maximum power dissipated

$$P_{\max} = I_0^2 R = \frac{45^2}{R^2}\times R = \frac{45^2}{R} = \frac{45^2}{900}\times 8 = 18.4W$$

11. (*c*) We know that, $I = \dfrac{V}{\sqrt{R^2+(X_L-X_C)^2}} = \dfrac{V}{Z}$

According to the diagram current increases as ω increases and we know that for a capacitive circuit,

$$I \alpha \frac{1}{X_c} \alpha\ \omega$$

Therefore for $\omega < \omega_r$, the circuit is mainly capacitive.

At $\omega = \omega r$, $X_L = X_C$

$$z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$z = R \qquad [\therefore X_L = X_C]$$

12. (*c*) When key is open current in the circuit

$$i_{rms} = \frac{15}{60} = \frac{1}{4}A$$

Therefore, $20 = i_{r.m.s.} \times L$

$$20 = \frac{1}{4}\times100\times L \Rightarrow L = 0.8\text{ H}$$

And $10 = i_{r.m.s.} \times C$

$$10 = \frac{1}{4}\cdot\frac{1}{100C} \Rightarrow C = 2.5\times10^{-4}\ F = 250\ \mu F$$

13. (*c*) Impedance of circuit, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\Rightarrow Z = \sqrt{(120)^2 + (10-100)^2} = 150\Omega$

So peak current, $I_0 = \dfrac{V_0}{Z} = \dfrac{30}{150} = 0.2\,A$

For restaurant frequency

$X_L = X_C$

$\omega L = \dfrac{1}{\omega C}$

$\omega^2 = \dfrac{1}{LC}$

$\omega = \dfrac{1}{\sqrt{LC}} = \dfrac{1}{\sqrt{10^{-1} \times 10^{-4}}} = \sqrt{10^5}$

$\because \omega = 2\pi f$

$\therefore f = \dfrac{\sqrt{10^5}}{2\pi} = 50\,Hz$

14. (*b*) Given, $X_L = 250\pi\Omega\, f = 50H_2$

At resonance, power is maximum

$\therefore X_L = X_C = \dfrac{1}{\omega C}$

$\Rightarrow C = \dfrac{1}{\omega X_L} = \dfrac{1}{(2\pi f)X_L}$

$= \dfrac{1}{2\pi \times 50 \times 250\pi} = \dfrac{1}{250 \times 100 \times \pi^2}$

$= 4\mu F$ $\quad [\pi^2 = 10]$

15. (*b*) For *DC* current

$H_{DC} = iR_1 t$ and *AC*

$H_{AC} = i_{\text{rms}}^2 R_2 t$

$\dfrac{H_{DC}}{H_{AC}} = \dfrac{i^2}{i_{\text{rms}}^2}\dfrac{R_1}{R_2} = \dfrac{(4)^2}{\left(\dfrac{4}{\sqrt{2}}\right)^2}\dfrac{3}{2} = 3:1$

16. (*a*) Power delivered = 1000 W

Voltage = 220 V

The current through the power line is, $I = \dfrac{1000}{220} = \dfrac{50}{11}A$

Power loss = $I^2R = \left(\dfrac{50}{11}\right)^2 \times 2 = 41.322W$

Efficiency

$= \dfrac{1000 \times 100}{1000 + I^2R} = \dfrac{1000 \times 100}{1000 + 41.322} \times 100 = 96\%$

17. (*c*) $E = Pt$

$= \dfrac{E^2}{Z^2}Rt = \dfrac{(24)^2}{60^2 + (8.33\pi - 2\pi)^2}(60)(60) = 518J.$

18. (*c*) Based on theory.

19. (*c*) Power equation of a transformer is given as

$e_1 i_1 \times \eta = e_2 i_2$

$\Rightarrow \quad 2300 \times 5 \times \dfrac{90}{100} = 230 \times i_2$

$\Rightarrow \quad i_2 = 45$ A

20. [11] For the current, $i(t) = A_0 \sin \omega t + B_0$

RMS value $= \sqrt{\dfrac{A_0^2}{2} + B_0^2}$

$I_{rms}^2 = \left(\dfrac{\sqrt{42}}{2}\right)^2 + 10^2 = 121$

$\Rightarrow I_{rms} = 11A$

21. [25] It is given that,

For a RLC circuit, R = 80 Ω, $X_L = 100\Omega$, $X_C = 40\Omega$.

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\sqrt{80^2 + (100-40)^2} = 100\Omega$

$i_0 = \dfrac{V_0}{Z} = \dfrac{2500}{100} = 25A$

22. [3872] At resonance $X_C = X_L$

$\Rightarrow \dfrac{1}{2\pi fC} = 2\pi fL$

$\Rightarrow C = \dfrac{1}{4\pi^2 f^2 L} = \dfrac{1}{4 \times \pi^2 \times 49 \times 10^6 \times 2 \times 10^{-6}}$

$\Rightarrow C = \dfrac{1}{3872}F$

$\therefore x = 3872$

23. [0] Given, $V_L = V_C = 2\,V_R$ [Resonance condition]

$\therefore V_L = 2 \times 220 = 440$ V

Current through the circuit, $I = \dfrac{220}{5} = 44A$

$\therefore V_L = I \times X_L = 440$ V

$\Rightarrow X_L = \dfrac{440}{44} = 10A$

$\Rightarrow L \times 2\pi \times 50 = 10$

$\Rightarrow L = \frac{10}{100\pi} = \frac{1}{10\pi} = \frac{100}{\pi}\text{mH}$

Comparing with $L = \frac{1}{K\pi}\text{mH}$, we get $K = 0.01$

24. [250] Bandwidth, $\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L}$

$\Rightarrow L = \frac{R}{\Delta\omega} = \frac{5}{20} = .25H = 250\text{ mH}$

25. [100] Capacitance, $C = 0.01 \times 10^{-6} \times 100 = 1 \times 10^{-6}\, F$

Frequency, $f = 0.5 \times 10^3\, s^{-1}$

For minimum impedance:

$x_L = x_C$

$\Rightarrow \omega^2 = \frac{1}{LC} \Rightarrow (2\pi f)^2 = \frac{1}{LC}$

$L = \frac{1}{4\pi^2 f^2 C} = \frac{1}{4\times(3.14)^2\times(0.5\times10^3)\times1\times10^{-6}}$

$= 10^{-1}\text{ H} = 100\text{ mH}$

26. [2000]

Quality factor

$= \frac{\omega L}{R} = \frac{2\times3.14\times10\times10^6\times2\times10^{-4}}{6.28} = 2000$

27. [125] At resonance, $\cos\phi = 1$

$\therefore P = V_{rms} I_{rms} \cos\phi$

$= V_{rms} I_{rms} = V_{rms} \times \frac{V_{rms}}{R} = \frac{V_{rms^2}}{R} = \frac{250^2}{5}$

$= 12500$ Watt $= 125 \times 10^2$ Watt

28. [500] Since, power dissipated is same, therefore,

$P_1 = P_2$

$\frac{220\times220}{5} = 220\times\frac{220}{z}\times\frac{5}{z}$

$z = R = 5$

Angular frequency,

$\omega = \frac{1}{\sqrt{LC}}$

$\omega = \frac{1}{\sqrt{0.1\times40\times10^{-6}}} = \frac{1}{2\times10^{-3}}$

$\omega = 500$ rad/s

29. [15] Impedance of the circuit

$Z = \sqrt{R^2 + (x_L - x_C)^2} = \sqrt{R^2 + \left(\omega_L - \frac{1}{\omega_C}\right)^2}$

$= \sqrt{(120)^2 + \left[100\times100\times10^{-3} - \frac{1}{100\times100\times10^{-6}}\right]^2}$

$= \sqrt{(120)^2 + (90)^2} = 150\Omega$

Current in the circuit,

$I = \frac{V_{r.m.s}}{Z} = \frac{\left(\frac{V_0}{\sqrt{2}}\right)}{Z} = \frac{20}{150} = \frac{2}{15}$

Now, Heat dissipated in the resistor $= ms\Delta T$

$I^2 R\Delta t = ms\Delta T$

$\Delta t = \frac{(ms\Delta T)}{I^2 \cdot R}$

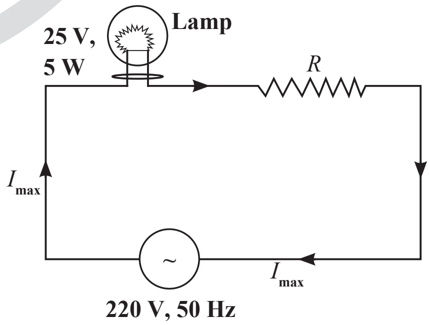
$= \frac{(2\times16)}{\left(\frac{2}{15}\right)^2\times120} = 15$ sec.

30. [975] For lamp, $P = \frac{V^2}{R}$

$\Rightarrow R = \frac{25\times25}{5} = 125\Omega$

From ohm's law

$I_{max} = \frac{V}{R} = \frac{25}{125} = \frac{1}{5}A$



$\Rightarrow I_{max} = \frac{220}{125+R} = \frac{1}{5} \quad \Rightarrow 1100 = 125 + R$

$R = 975\Omega$